IPEM Group of Institutions, Ghaziabad
Department of Computer Application
Assignment – I

Course: Feature Engineering | **Programme:** BCA III Semester
Assignment Date: 10-08-2026 | **Submission Date:** 12-08-2026

Instructions:
1. Answer all questions in your own words.
2. Write neat and well-explained answers with suitable examples wherever required.
3. Draw diagrams or tables wherever applicable.
4. Submit the assignment on or before the submission date.
5. Late submissions may not be accepted.

Section A – Short Answer Questions
1. Define Feature Engineering. Why is it important in Machine Learning?
2. Differentiate between data and features with suitable examples.
3. Explain the difference between numerical and categorical features.
4. Distinguish between discrete and continuous data with examples.
5. What is the difference between interval and ratio scales? Give one example of each.
6. Define ordinal data. How is it different from nominal (categorical) data?
7. What are missing values? Mention any four reasons why missing values occur in real-world datasets.
8. Explain the terms MCAR, MAR, and MNAR.
9. What is data cleaning? Why is it an essential step before building a Machine Learning model?
10. Mention any five common data cleaning techniques used during data preprocessing.

Section B – Long Answer Questions
1. Explain the importance of Feature Engineering in Machine Learning. Discuss how good features improve model performance with suitable examples.
2. Classify data into Numerical, Categorical, Ordinal, Discrete, Continuous, Interval, and Ratio data. Explain each type with appropriate real-life examples.
3. What is missing data? Explain the different types of missing data (MCAR, MAR, and MNAR) with suitable examples. Also discuss the challenges caused by missing values in Machine Learning.
4. Explain the complete process of data cleaning. Discuss various data quality issues and the techniques used to handle missing values, duplicate records, inconsistent data, and incorrect data entries.
5. A college has collected student data containing missing attendance, incorrect ages, duplicate records, and inconsistent department names. As a data analyst, explain how you would identify and clean the dataset before using it for Machine Learning. Illustrate your answer with suitable examples.